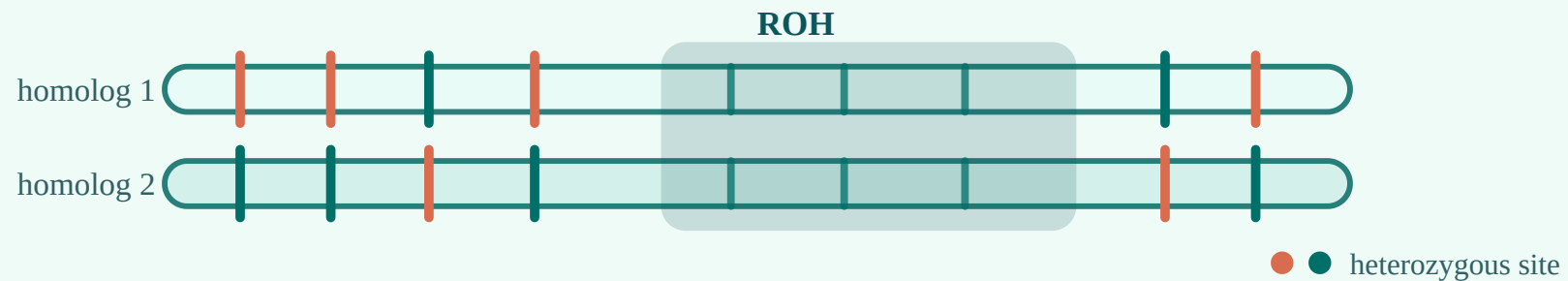


Genetic Diversity, Heterozygosity, and ROH



Plan for This Session

1

What genetic
diversity means

2

Heterozygosity

3

Calculating
heterozygosity

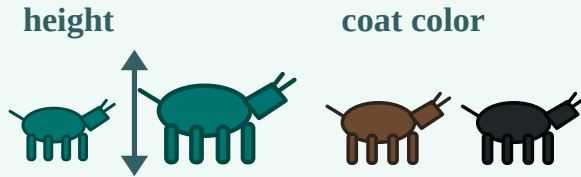
4

Inbreeding and
ROH

The exercise then turns these ideas into counts, windows, ROH calls, and plots.

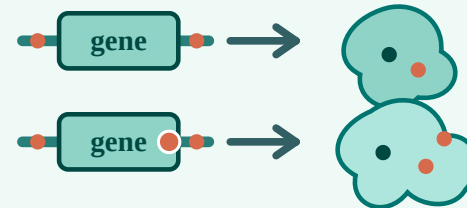
Diversity We Can See and Diversity We Cannot

Phenotypic variation



Traits, morphology, behaviour, survival, reproduction

Functional variation



Immune genes, fertility, disease resistance, adaptation

Neutral variation



Genome-wide variation shaped mostly by demography

Observed traits are what we often see first

can reflect

Functional variants affect some biological outcomes

within

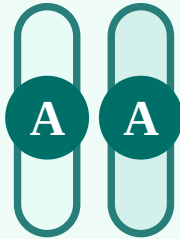
Neutral variation tracks demography across the genome

Why Genetic Diversity Matters

- **Short term:** fertility, survival, disease resistance, inbreeding depression
- **Long term:** raw material for adaptation to future change
- **Management:** monitoring, translocations, genetic rescue, captive breeding

Heterozygosity at One Site

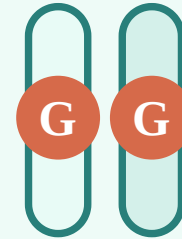
A/A



A/G



G/G



At one site, heterozygosity is simple: does the individual carry two different alleles?

Quick check: which genotype is heterozygous, and which are homozygous?

Why Heterozygosity Measures Diversity

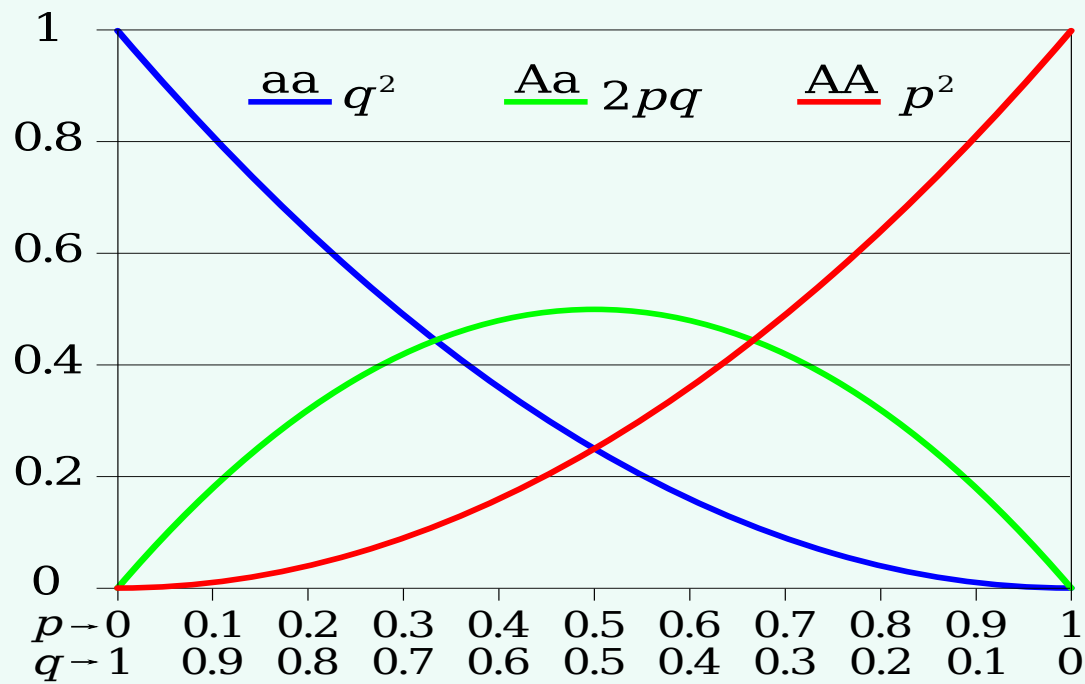


Figure: Johnuniq, Wikimedia Commons, CC BY-SA 3.0 / GFDL.

- Fixed sites cannot be heterozygous
- Rare alleles create little heterozygosity
- Sites with balanced allele frequencies contribute most

The heterozygote curve, $2pq$, is highest when both alleles are common.

Heterozygosity Across a Genome

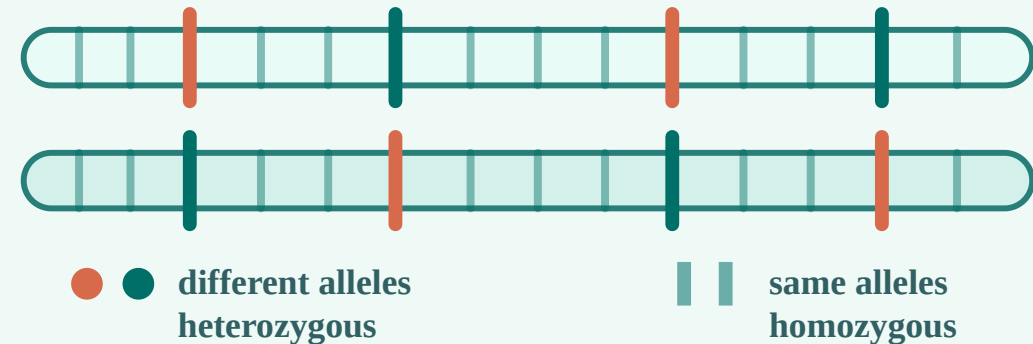
Individual heterozygosity

Proportion of callable sites that are heterozygous in one individual

Across samples

Compare the same kind of estimate among individuals, populations, or time points

Count heterozygous sites across callable DNA



What Changes Heterozygosity?

Mutation



Adds new alleles, usually slowly at population-genomic timescales

Gene flow



Can increase H when migrants bring different alleles

Drift



Random allele loss; strongest in small populations

Bottlenecks



Sharp loss of variation, especially rare alleles

Inbreeding



Reduces individual H by pairing related chromosome copies

Selection



Can reduce diversity, but balancing selection can maintain or increase local H

Pause: which forces add variation, which remove it, and which can do either?

Simple Formula, Hard Denominator

Conceptually:

H = het sites / callable sites

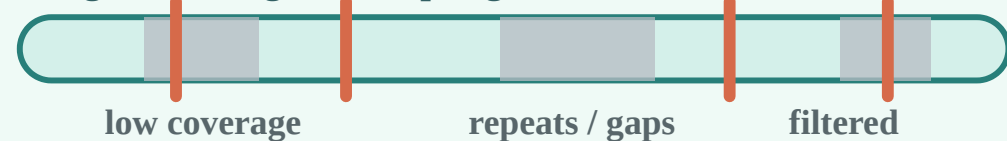
Practically ask:

- Which sites were callable?
- Were invariant sites retained?
- Do some samples or groups have many more missing genotypes?

Pause: why not use genome size as the denominator?

The denominator is a decision

full genome length is tempting



callable space after filtering



Only callable sites belong in the denominator.

SNP Heterozygosity vs Genome-Wide Heterozygosity

SNP heterozygosity

- polymorphic sites only
- useful within matched analyses
- biased as total genomic diversity
- can strongly inflate the number

Genome-wide heterozygosity

- callable autosomal sites
- invariant + polymorphic sites
- closer to total genomic diversity
- depends on a defensible callable denominator

Comparing Heterozygosity

Individuals

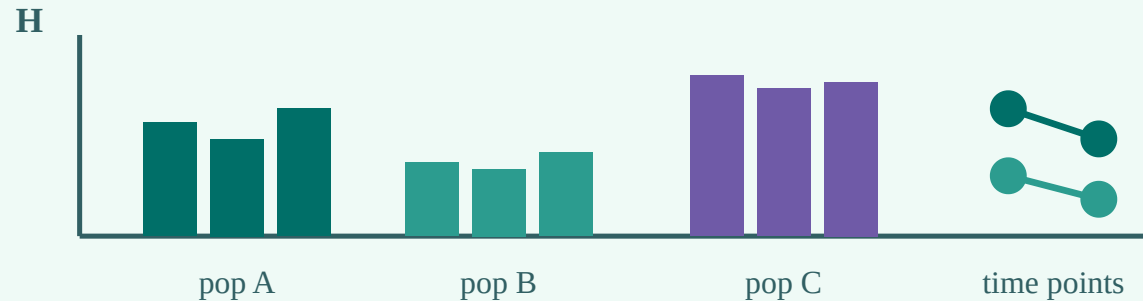
Which individuals have unusually low H within the same population?

Populations

Which populations have lost more variation?

Time points

Has diversity changed through time, including historical samples?

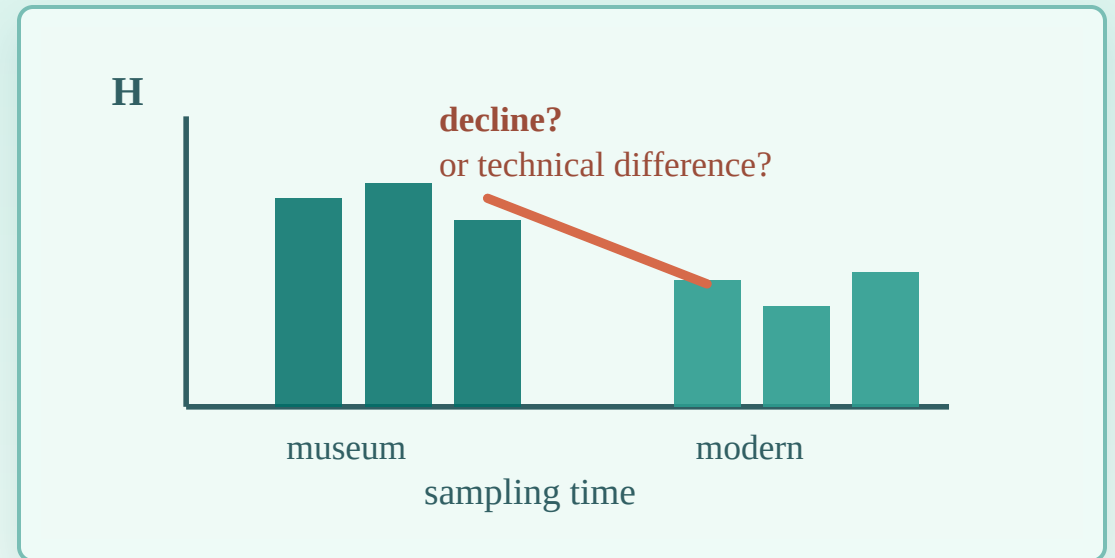


Discuss:

Which comparison is easiest to trust, and which is most vulnerable to technical differences?

Temporal Comparisons Can Be Powerful

- Museum or historical samples can anchor an earlier baseline
- Modern resampling can test whether diversity is still changing
- Age, degradation, coverage, and missingness can mimic biological change



What Makes a Comparison Fair?

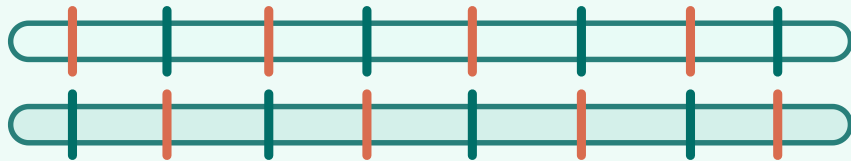
- Matched genomic space, or differences explicitly accounted for
- Similar filtering and missing-data thresholds
- Similar sequencing depth or genotype uncertainty handling
- Comparable variant discovery strategy
- Individual estimates shown, not only population means

The point is to avoid mistaking data-processing differences for biological differences.

What Heterozygosity Can Hide

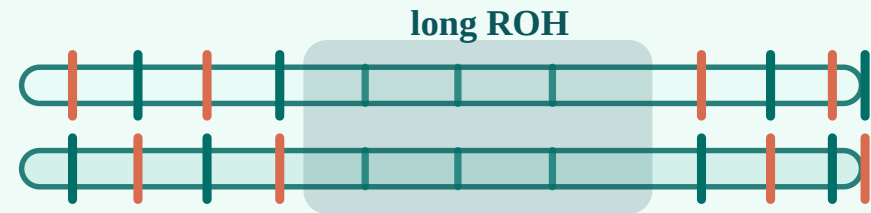
Same genome-wide heterozygosity. Different genome.

Individual A



8 heterozygous sites, spread out

Individual B



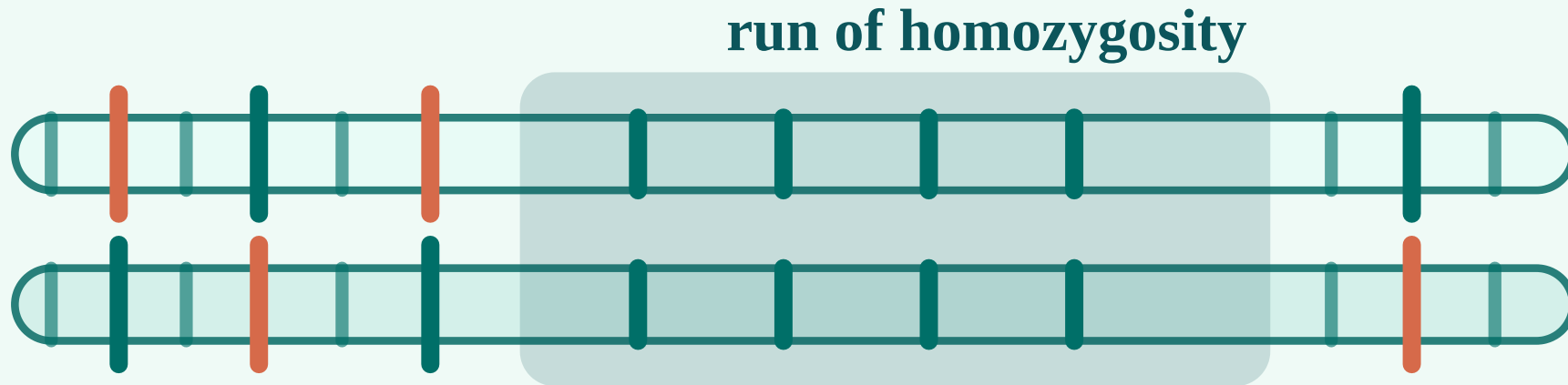
8 heterozygous sites, plus a long homozygous block

Question: what could cause one individual to have a long stretch with no heterozygous sites?

Runs of Homozygosity

Biological idea: the two chromosome copies are identical by descent across a stretch of genome.

Data reality: called ROH are long tracts with few or no observed heterozygous genotypes.



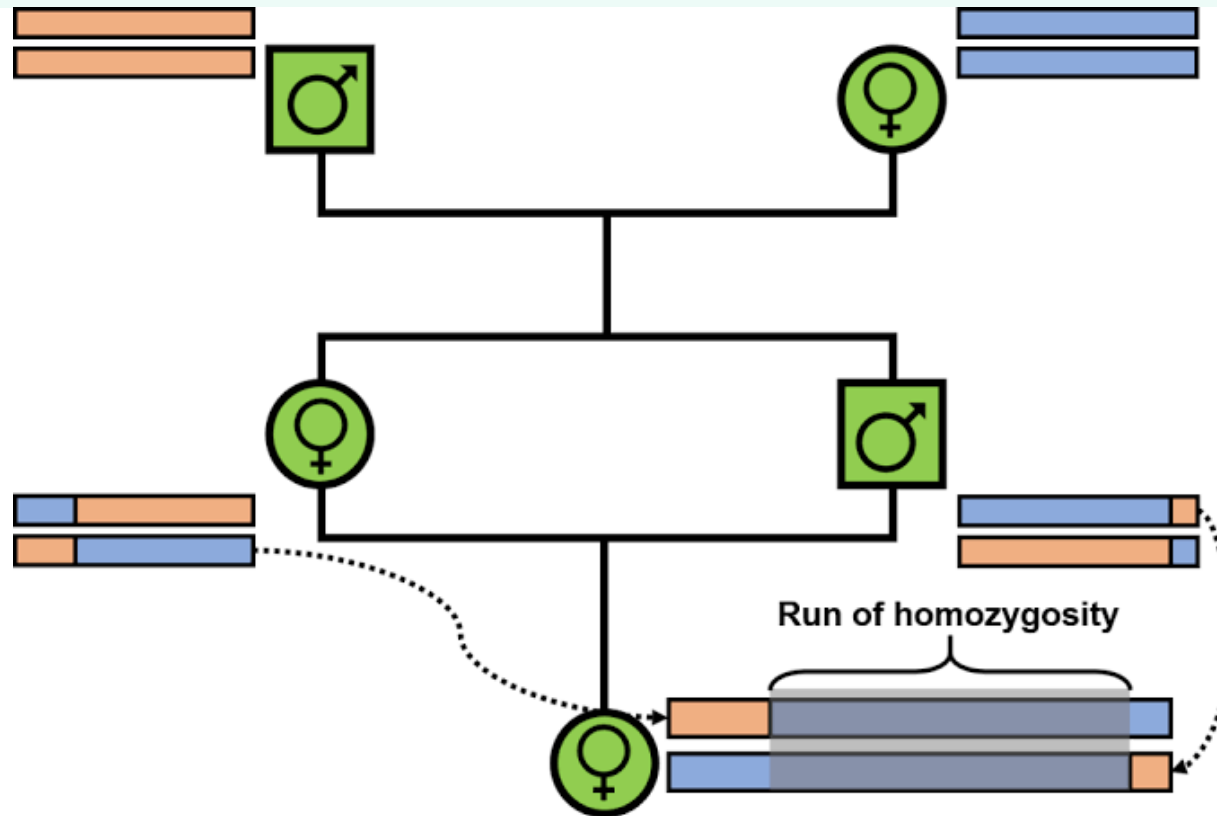
No true heterozygous sites inside autozygous ROH; homozygous sites occur outside too

Inbreeding Starts with Shared Ancestors

- Inbreeding occurs when parents are related
- Related parents may pass on the same ancestral segment
- Offspring may inherit matching copies from both sides

Pedigrees describe the expectation; genomes show which segments were actually inherited.

From Pedigree to ROH



Identity by Descent and Autozygosity

- **Identity by descent:** two alleles trace back to the same ancestral copy
- **Autozygosity:** homozygosity because of identity by descent
- **Pedigree F:** expected autozygous fraction

Expected F of offspring

Parents are...	F
unrelated	0
first cousins	1/16
half-sibs / avuncular	1/8
full sibs	1/4
parent-offspring	1/4

Observed vs Expected H

Observed H_O

Fraction of genotypes that are heterozygous

Expected H_E

Predicted from allele frequencies

Under HWE, H_E is the expected heterozygote frequency.

H_O vs H_E can flag inbreeding, structure, or technical issues.

For allele frequencies p and q :

$$H_E = 2pq$$

More generally:

$$H_E = 1 - \sum_i p_i^2$$

Inbreeding as a Heterozygote Deficit

Inbreeding makes the two allele copies within an individual less independent.

- H_O decreases relative to H_E
- homozygous genotypes increase
- the signal is often summarized by F

$$H_O = H_E(1 - F)$$

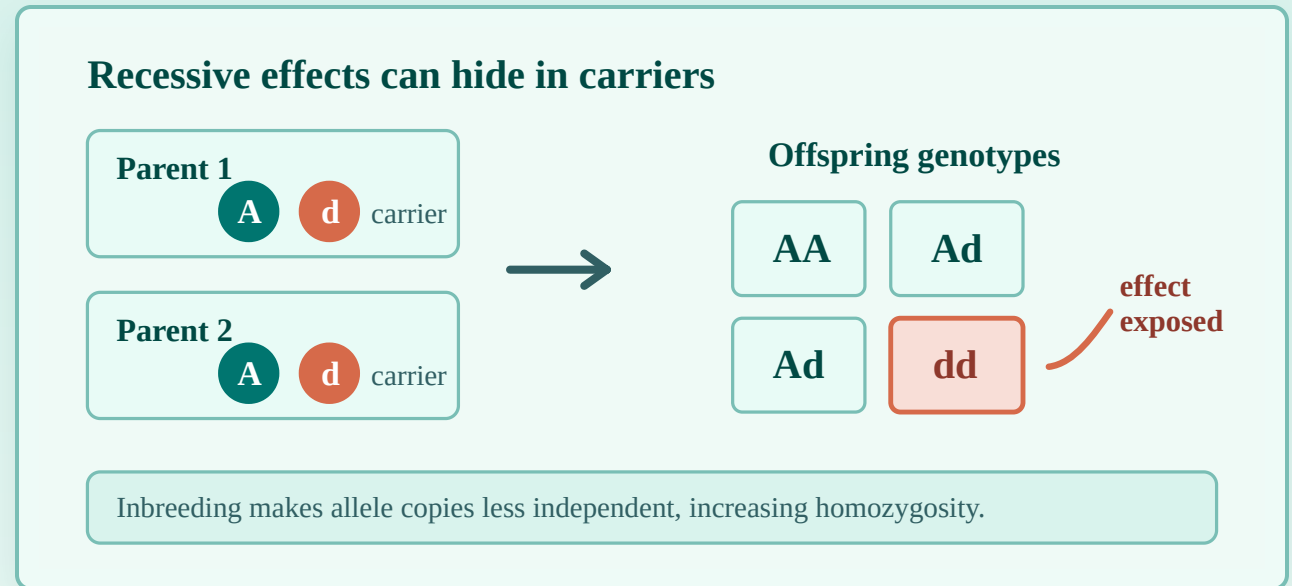
$$F = 1 - \frac{H_O}{H_E}$$

$F > 0$ means fewer heterozygotes than expected under random mating.

Pause: what else besides inbreeding can make H_O lower than H_E ?

Why Inbreeding Can Reduce Fitness

- Inbreeding increases homozygosity across the genome
- Recessive deleterious effects can be hidden in heterozygotes
- Homozygosity can expose those effects
- The fitness loss is **inbreeding depression**
- Not all homozygosity is harmful



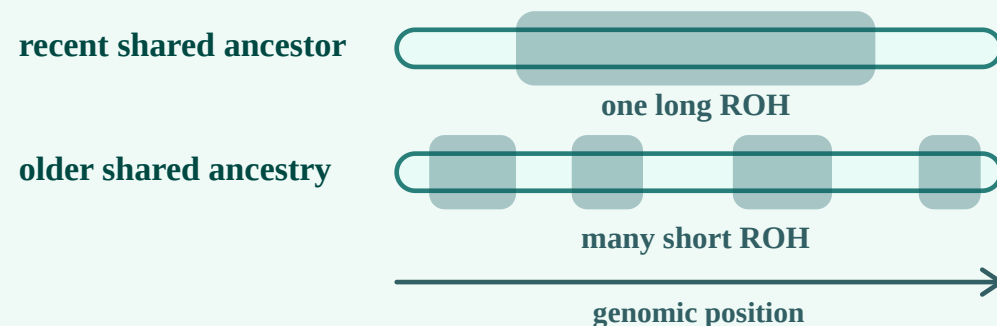
ROH Length Carries Time Information

Long ROH

- recent shared ancestor
- fewer recombination events
- often recent close-kin mating

Short ROH

- older shared ancestry
- broken down by recombination
- often older bottleneck or long-term small N_e

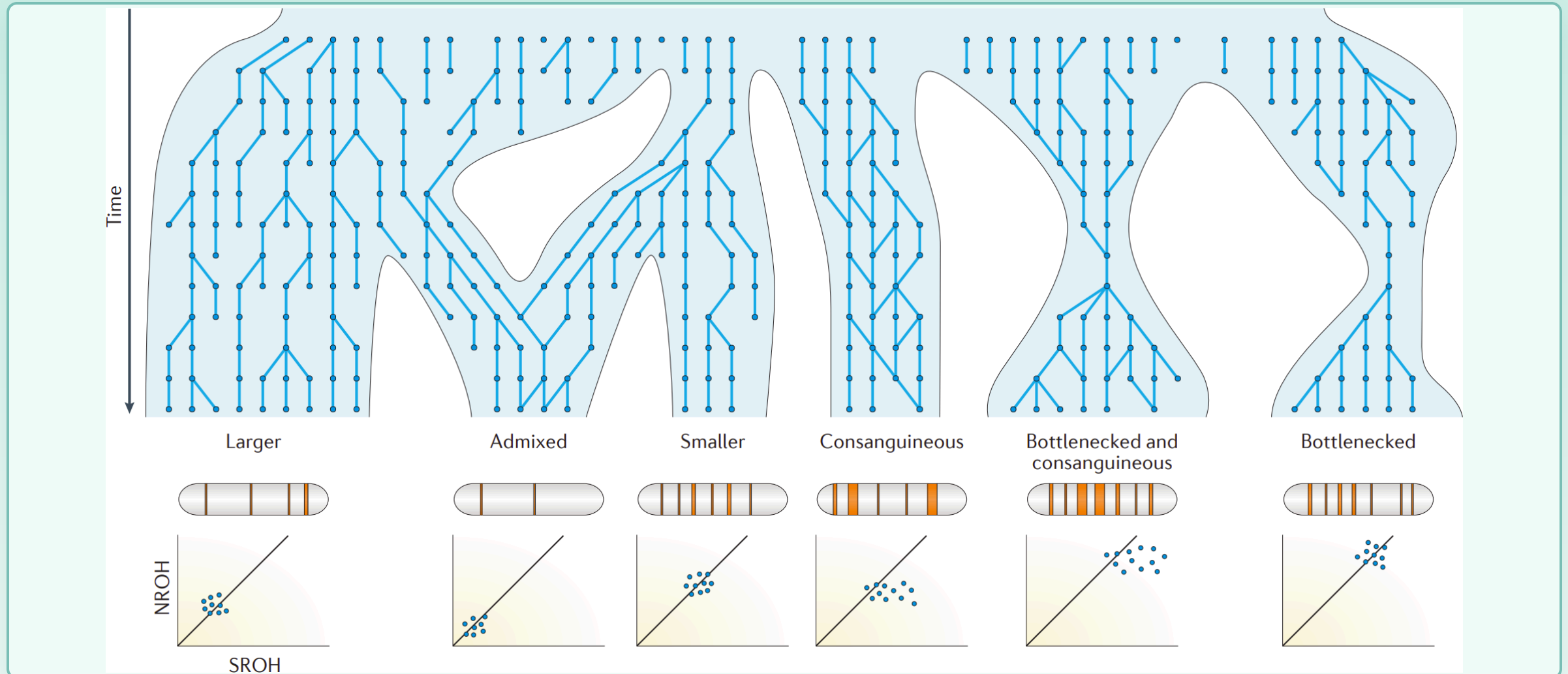


ROH Are Common; Pattern Matters

- Some ROH are expected, even in broadly outbred populations
- The **length distribution** separates recent and older shared ancestry
- The **total ROH length** gives a genome-wide autozygosity summary
- The **genomic location** can flag repeated regions across individuals

ROH are strongest when interpreted as a pattern: length classes, total length, and where they occur.

Demographic Origins of ROH



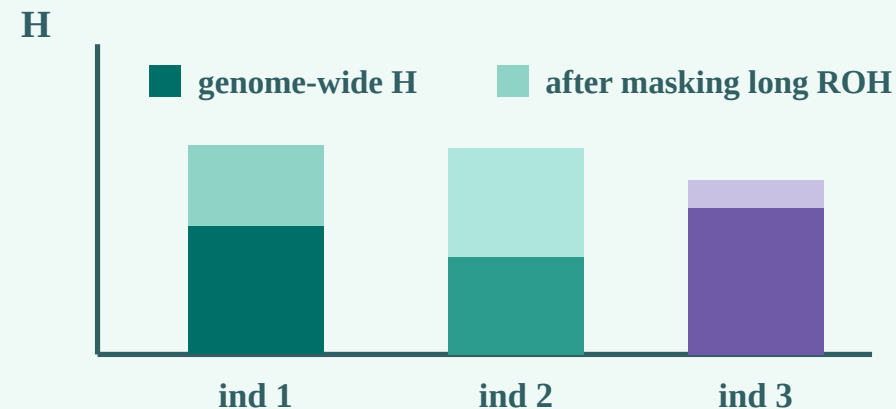
Source: Ceballos et al. 2018, Nature Reviews Genetics, doi:10.1038/nrg.2017.109.

Heterozygosity Outside Long ROH

Sometimes it is useful to estimate heterozygosity after masking long ROH.

Prediction: after masking long ROH, should heterozygosity go up, go down, or stay the same?

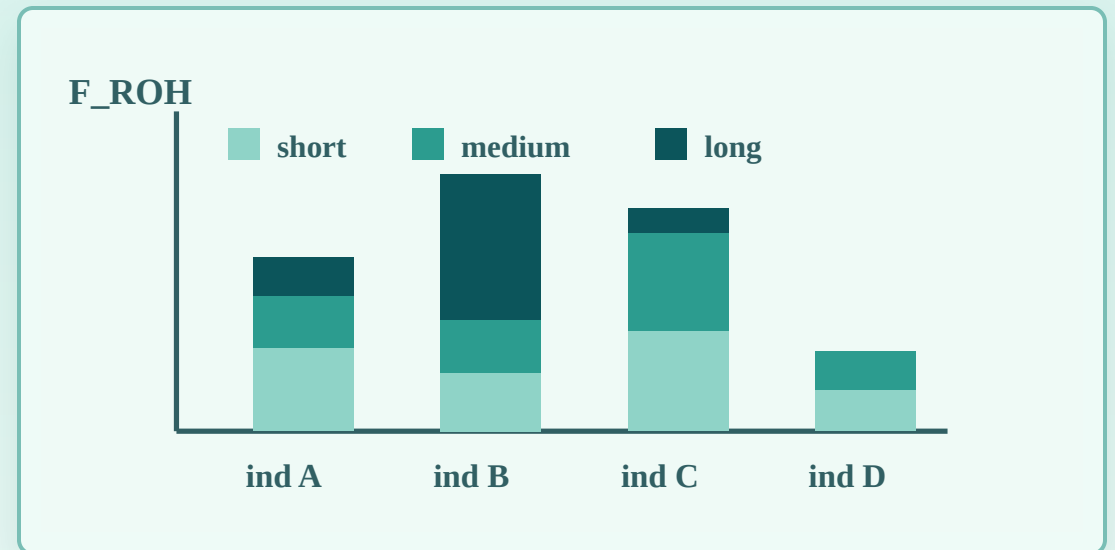
This can help separate background genomic diversity and recent close inbreeding.



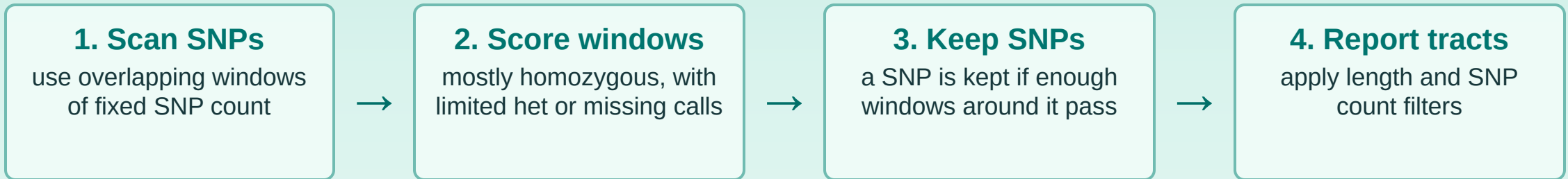
Summarizing ROH

- number of ROH
- total ROH length
- longest ROH
- ROH length classes

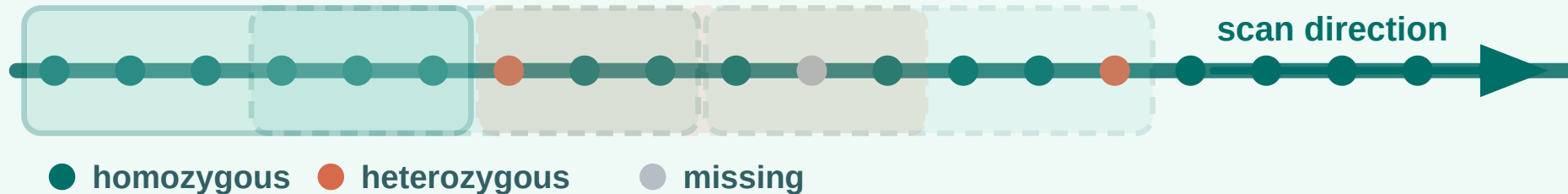
$$F_{ROH} = \frac{\text{total length of ROH}}{\text{genome length considered}}$$



How PLINK-Style ROH Calling Works



Overlapping windows scan across markers



PLINK's `--homozyg` call set is a heuristic scan of marker data, so parameter choices and sanity checks matter.

ROH Calls Depend on Parameters

- minimum ROH length
- minimum number of SNPs
- allowed heterozygous calls
- allowed missingness
- parameter choices often need a few sensible trials

Check inferred ROH against heterozygosity plots, known relatedness, and population history.

Take-Home Messages

1. Heterozygosity tells us how much variation is present.
2. Filtering defines what we are actually measuring.
3. ROH tell us how homozygosity is arranged within individuals.
4. Long ROH usually point to recent shared ancestry.

Time for Exercises!